Abdulhafiz Muzyin

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EDUCATION

Georgia Institute of Technology **Atlanta, GA**
Bachelor of Science in Computer Engineering | GPA: 3.91/4.00 *Expected May 2028*
Relevant Coursework: Circuit Analysis, Digital System Design, HW/SW Systems Programming, Data Structures & Algorithms
Honors: Phi Theta Kappa; CLACEC Coding Competition Winner (2025); Dean's Award Winner; STEM Scholar of Excellence

TECHNICAL SKILLS

Languages & HDL: C, C++, SystemVerilog, Python, RISC-V Assembly, MATLAB, Java
Hardware & Verification: RTL Design, FSMs, FPGA Prototyping, Functional Verification, Constrained-Random Testing, SVA, Testbenches, Logic Synthesis, Place & Route, Static Timing Analysis (STA)
Embedded & Electrical: Embedded C/C++, Microcontrollers, PWM Motor Control, PID Control, Sensor Integration, PCB Design
EDA & Tools: Cadence Xcelium, SimVision, Innovus, Pegasus; Synopsys Design Compiler; Intel Quartus Prime; AutoCAD; Git; Linux

ENGINEERING EXPERIENCE

Digital Design & Verification Engineer | Silicon Jackets, Georgia Tech **Aug 2026 – Present**
- Designed a 32-bit single-cycle RISC-V (RV32I) CPU in SystemVerilog with 9 instructions and load-stall control.
- Verified RTL using 13 directed tests, 20 constrained-random programs, and 300 randomized memory-port transactions with a reference-model scoreboard and SVA; debugged failures in SimVision waveforms.
- Executed logic synthesis, floorplanning, and place-and-route in SkyWater 130nm using Synopsys Design Compiler and Cadence Innovus; reduced total negative slack (TNS) by 32% through clock gating and cell optimization.

Firmware & Electrical Engineer | RoboJackets, Georgia Tech **Aug 2026 – Present**
- Developed embedded C/C++ firmware for an autonomous Micromouse, using Flood Fill pathfinding to solve 16×16 mazes.
- Implemented PID motor control and PWM speed regulation with encoder feedback and IR sensors for wall detection.
- Integrated and debugged sensor circuits, motor drivers, and MCU interfaces, validating HW/SW through maze tests.

UX Researcher | Global Assistive Technology Innovation VIP, Georgia Tech **Aug 2026 – Present**
- Interviewed stakeholders in the U.S. and Rwanda and translated findings into requirements for a collaboration platform.
- Collaborated with a multidisciplinary engineering team to define features and user workflows for the Confluence platform.

Manufacturing & Process Improvement Intern | Al-Nur Industrial Group P.L.C **Dec 2025 – Jan 2026**
Addis Ababa, Ethiopia
- Optimized the cable winding-to-packaging workflow, cutting handling time by 6 seconds per roll.
- Designed and integrated a QR-code verification system for 2.5 mm electrical cables to reduce counterfeit risk.
- Performed electrical cable quality inspection and supported equipment commissioning and operator training.

TECHNICAL PROJECTS

FPGA-Based Digital Access Controller | SystemVerilog, Intel DE10-Standard, Quartus Prime **2026**
- Designed a password-protected access controller in SystemVerilog using FSMs and sequential logic to control lock states.
- Integrated FPGA switches, push buttons, and LEDs with RTL logic for password entry, reset, and lock/unlock feedback.
- Verified FSM transitions, valid/invalid sequences, and reset behavior through simulation and on-board testing.

Obstacle-Aware Autonomous Robotic Vehicle | Embedded C++, Arduino, Ultrasonic Sensors **2025**
- Built a 4WD autonomous vehicle on Arduino Uno with a servo-mounted HC-SR04 ultrasonic sensor for 180° scanning.
- Implemented collision avoidance (20 cm threshold) and PWM motor control through an L298N H-bridge driver.
- Isolated motor-induced electrical noise from microcontroller circuitry, reducing system reset errors by 38%.

LEADERSHIP

Co-Founder | Egnaw Le Egnaw (Us for Us) Foundation **Mar 2025 – Present**
- Co-founded a nonprofit that raised $6,350+ (2025–2026) for under-resourced rural schools in Ethiopia.
- Renovated 4 classrooms and provided 272 textbooks, computers, and Wi-Fi, supporting a 14% rise in exam pass rates.

ADDITIONAL

Languages: English (Fluent), Amharic (Fluent), Spanish (Working Proficiency)
Organizations: ColorStack, National Society of Black Engineers (NSBE), GTEESA